

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

End-Semester Exam: Closed Book / Notes / Everything

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Basic Matrix Identity:

$$(P^{-1} + B^{\top} R^{-1} B)^{-1} B^{\top} R^{-1} = P B^{\top} (B P B^{\top} + R)^{-1}$$

KL divergence between Gaussians $G(\cdot; \mu_1, \sigma_1^2)$ and $G(\cdot; \mu_2, \sigma_2^2)$ (in that order of arguments) is

$$0.5((\mu_2 - \mu_1)^2 / \sigma_2^2 + \sigma_1^2 / \sigma_2^2 - \log(\sigma_1^2 / \sigma_2^2) - 1)$$

1. [35 points]

Your goal is to train a variational autoencoder (VAE) for RGB microscopy images of size 256×256 pixels.

- (10 points) Describe an appropriate convolutional neural-net architecture for the encoder and the decoder within the VAE. Describe each layer in detail. Describe kernel sizes clearly.

- (10 points) Mathematically derive the objective function for VAE training involving the neural-net parameters θ for the encoder and ϕ for the decoder, given training data $\{X_n\}_{n=1}^N$.

- (5 points) How does the VAE objective function differ from the (log) likelihood function of the training dataset ? Give a mathematical expression.

- (5 points) Specify the mathematical expression for the reparametrization for the latent variable, during VAE training.

- (5 points) Give a step-by-step implementation algorithm for VAE learning relying on gradient descent using a backpropagation library (assume this is already available to you).

2. [40 points]

Consider the regression of the dependent variables corresponding to a person's age, i.e., $\{y_n \in \mathbb{R}\}_{n=1}^N$, using the corresponding independent variables (features) that extracted from dental CT images, i.e., $\{z_n \in \mathbb{R}^D\}_{n=1}^N$, where $1 \ll D < \infty$.

- (5 points) Derive a Bayesian formulation for estimating a linear regressor using ridge regression, i.e., specify the **likelihood** function, **prior** distribution, **posterior** distribution, and the **objective function**.

- (5 points) Derive a **non-iterative** implementable algorithm for Bayesian estimation (as described above) of a linear regressor using ridge regression.

- (10 points) Derive an implementable algorithm for estimating a linear regressor when you know that there are severe data-collection errors in the values of the dependent variables in the training dataset.

- (10 points) Derive a **non-iterative** implementable algorithm for estimating a **kernel** ridge regressor using a radial-basis-function kernel.

- (10 points) Derive an implementable algorithm for estimating a **kernel** regressor with a radial-basis-function kernel when you know that only a few independent variables are sufficient for good prediction.

3. [20 points]

Consider a problem of reconstructing a spatio-temporal cardiac MRI image X (complex-valued intensities) from the acquired data matrix Z (complex-valued components), where the intensities vary smoothly over time.

- (8 points: 4 + 4) Formulate a Bayesian image reconstruction problem, specifying and justifying (i) the likelihood model and (ii) the prior model.

- (5 points) Suppose you want to acquire only a subset of the data Z , undersampling the data collection within each timeframe, in order to speedup the data acquisition process. Design a data undersampling scheme in k -space that is (i) practical, (ii) effective, and, at the same time, (iii) amenable to efficient reconstruction algorithms (as per the next sub-question).

- (12 points: 4 + 8) For the aforementioned scenario, (i) reformulate the reconstruction problem and (ii) describe an implementable algorithm for the image reconstruction using gradient-based optimization and deriving all gradients needed for optimization.

4. [25 points]

- (8 points) Consider a dataset of N points in D dimensions, i.e., $\{x_n \in \mathbb{R}^D\}_{n=1}^N$. Consider $N \gg D$. Consider that the dataset spans a subspace of $\mathbb{R}^D$ having dimension $E < D$. Prove mathematically that the Frechet mean, using the Euclidean distance metric in $\mathbb{R}^D$, must lie in the span of the data.

- (7 points) Prove mathematically that every *mode of variation* must be representable as a vector lying in the span of the data.

- (10 points: 5+ 5) Consider an application where all measured data (e.g., a time series) lies on a hypersphere, and the distance between two data points must be the length of the shortest path, between those two data, on the hypersphere. Consider a dataset of M points $\{y_m\}_{m=1}^M$, with each y_m lying on a hypersphere in D dimensions having center at the origin and radius 1. Design a mathematical formulation to estimate the Frechet mean for this spherical data, that also lies on the hypersphere. Design an implementable algorithm to compute the Frechet-mean estimate.

5. [20 points]

Consider brain-MRI segmentation (into 3 segments: white matter, gray matter, cerebrospinal fluid) using a Gaussian mixture model (GMM), expectation maximization (EM), and an atlas-based prior for the segments (without any MRF-based prior). To employ the atlas-based prior, assume that you have the brain-MRI template image and the associated probability maps for the three segments.

- (10 points) Formulate the $Q(\cdot)$ function, and specify how you can compute each term within that function exactly.

- (10 points) Clearly describe an algorithm to estimate the segmentation of an arbitrary subject's brain MRI, using atlas-based priors.